Question

It is known that $[XeF][AsF_6]$ reacts with H_2O in anhydrous HF according to the following equilibrium:

$$[XeF][AsF_6] + H_2O + HF \rightleftharpoons XeF_2 + [H_3O][AsF_6]$$

When equimolar amounts of $[XeF][AsF_6]$ and H_2O react in anhydrous HF at −78 °C, the adduct $[H_3O][AsF_6] \cdot 2XeF_2$ is initially formed. Subsequently, $[XeF][AsF_6]$ reacts with free XeF_2 in a 1:1 ratio to yield compound $\mathbf{A}$. This adduct further reacts with free XeF_2 to form a 1:1 salt $\mathbf{B}$. The anion of $\mathbf{B}$ is identical to that of $\mathbf{A}$, while its cation contains oxygen with a coordination number of 2. The mass fraction of Xe in $\mathbf{B}$ is approximately 60%. The cation of $\mathbf{B}$ is obtained from the cation of $\mathbf{A}$ by reacting with the electrically neutral species $\mathbf{C}$, a process involving a single proton transfer.

Which of the following statements is correct:

A. The cation of $\mathbf{A}$ is linear in shape.

B. The relative molecular mass of $\mathbf{A}$ is smaller than that of $\mathbf{C}$.

C. The cations of $\mathbf{B}$ are triangular in shape.

D. The chemical reaction equation for the formation of $\mathbf{B}$ (with coefficients in the simplest whole-number ratio), where the sum of the coefficients on the right side of the equation is 3.

E. $\mathbf{C}$ contains 3 types of elements.

F. The number of atoms contained in $\mathbf{C}$ is 4

Answer

Correct Answer: **F**

Detailed Explanation

The formation of $\mathbf{A}$ is straightforward. Since it is a 1:1 reaction, $\mathrm{XeF_2}$ can only insert into $\mathrm{XeF^+}$ to yield $\mathrm{Xe_2F_3^+}$, which is the cation of $\mathbf{A}$. According to VSEPR theory, one F atom bridges the two Xe atoms and adopts $\mathrm{sp^3}$ hybridization, resulting in a trigonal shape rather than a linear geometry. Therefore, option A is incorrect.

CHECKPOINT 2 PTS

The cation of $\mathbf{A}$ is $\mathrm{Xe_2F_3^+}$ with a trigonal shape

The anion of $\mathbf{B}$ is the same as that of $\mathbf{A}$, both being hexafluoroarsenate anions. The cation contains oxygen with a coordination number of 2, suggesting it is composed of $\mathrm{Xe, O,}$ and F. Combining the mass fraction, the chemical formula of $\mathbf{B}$'s cation is determined to be $\mathrm{Xe_3OF_3^+}$.

CHECKPOINT 2 PTS

The chemical formula of $\mathbf{B}$'s cation is $\mathrm{Xe_3OF_3^+}$

Considering its structure, oxygen cannot be a terminal atom and must act as a bridging atom. Based on the structure of $\mathrm{Xe_2F_3^+}$, $\mathrm{Xe_3OF_3^+}$ can be derived by replacing a terminal fluorine atom in $\mathrm{Xe_2F_3^+}$ with an oxygen atom and then attaching a XeF unit, described as $\mathrm{F-Xe-F-Xe-O-Xe-F}$. According to hybridization theory, it adopts a bent shape, making option C incorrect.

CHECKPOINT 1 PTS

$Xe_3OF_3^+$ has a bent shape

Thus, the reaction for forming $\mathbf{B}$ can be written as:

$$3XeF_2 + [H_3O][AsF_6] = [Xe_3OF_3^+][AsF_6] + 3HF$$

The sum of the coefficients on the right side of the equation is 4, so option D is incorrect.

$Xe_3OF_3^+$ is generated from $Xe_2F_3^+$ and $\mathbf{C}$ via a proton transfer process. Given that $Xe_3OF_3^+$ is equivalent to replacing a terminal fluorine atom in $Xe_2F_3^+$ with an oxygen atom and attaching a XeF unit, the displaced fluorine atom likely combines with the proton. Therefore, the structure of $\mathbf{C}$ is clearly $HOXeF$, containing four elements and four atoms. Thus, option F is correct, while options E and C are incorrect.

CHECKPOINT 1 PTS

The structure of $\mathbf{C}$ is $HOXeF$